A Holistic Approach to Diabetes Management: Integrating Sentiment Analysis with Clinical and Behavioral Data

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Abstract. Effective diabetes management requires understanding both physiological metrics and the patient's psychological state. However, current systems analyze these in isolation; clinical tools overlook patient sentiment, while sentiment analysis models are disconnected from physiological data. The paper proposes a novel integrative framework to bridge this gap and demonstrate a system that fuses two distinct data streams: (1) physiological and behavioral data (e.g., Glucose, BMI, HbA1c) from a diabetics dataset, and (2) textual emotion data from the GoEmotions dataset. The proposed methodology employs a sentiment modeling module, implemented using an Embedded Deep Neural Network (E-DNN), to classify text emotions. The sentiment vector is then concatenated with the clinical features in a fusion layer to create a unified, holistic emotional-health profile. The sentiment component is evaluated on a 20-sample test set, achieving an overall accuracy of 75.00% with the proposed E-DNN. The primary contribution is the framework itself, which addresses the critical data-silo limitation of existing research. By connecting emotional states to physiological data, The proposed multi-modal approach provides a richer, more contextualized understanding of patient well-being, paving the way for more personalized and effective diabetes care.

Keywords: Data Fusion, Diabetes Management, E-DNN, Holistic Health, Multi-modal Data, Sentiment Analysis.

1 Introduction

Diabetes mellitus is a very serious chronic metabolic disorder that affects the general metabolic system of the body. It affects the people of the world and it is estimated that the number will rise to 643 million by 2030. The disease, which is still a cause of death among the population, requires a very careful and thorough monitoring of the patient's physical health in order to overcome the complications resulting from it. At the same time, a demand has arisen for the development of sentiment analysis techniques. The application of natural

language processing to discover the emotional tone from the written text has been considered an influential means in the healthcare sector, thus facilitating the extraction of patient emotions and opinions not only from the direct conversation with the doctor but also from the online resources. These signals of the patient's situation imply that combining physiological data with emotional context provides a broader insight into the patient's health condition.

The International Diabetes Federation, estimates that in 2021 about 537 million adults (age 20–79) were suffering from diabetes. The number of diabetic patients is expected to rise by 2030 to 643 million [2]. The disease has been the cause of approximately 6.7 million deaths in 2021. These figures vividly reflect the urgency for improved diabetes care, as stipulated by the United Nations' Sustainable Development Goal 3 which requires a one-third reduction of premature mortality from non-communicable diseases (including diabetes) by 2030 [3]. Continuous glucose monitoring (CGM) systems and wearable sensors have made it possible to obtain physiological data in real time from patients and to transmit it to healthcare providers. Meanwhile, machine learning and artificial intelligence are progressively being utilized in the analysis of healthcare data to identify the risk. At the same time, the latest advanced sentiment analysis methods incorporate highly sophisticated NLP (mostly deep learning or large language models) for the interpretation of the patient-generated text [1].

On the other hand, extant solutions are fraught with significant drawbacks. The majority of diabetic management systems redesigns that physiologic or behavioral data only be taken into account, emotional context being ignored even if considered jointly. Similarly, sentiment analysis in Health seldom merges with clinical metrics. This fragmentation means that, for instance, changes in stress and mood of a diabetic person are not traced to their glucose patterns that, thus, can surely leave some care or patient engagement opportunities unnoticed. The convergence of behavioral, physiological, and emotional data imposes considerable obstacle. Data- heterogeneous as they are- a fusion method is needed to integrate the continuous time-series from sensors with the unstructured text. While data standardization, feature alignment, and shortage of multimodal training data are identified as the key barriers to this fusion process, on the sentiment side, creation of usergenerated text that is not free from noise and the presence of biases (e.g., sarcasm, cultural differences) that hamper the interpretability of the text, and the need for privacy make the problem even more intractable [1]. The present work proposes a novel integrative framework that combines sentiment analysis with physiological and behavioral diabetes data. The approach involves machine learning models that accept sensor measurements and sentiment indicators as inputs. The system, by integrating emotional and clinical data, can offer personalized insights that cannot be gleaned from any single data source alone. The main contributions are as follows:

1. A unified framework that combines physiological and diabetes data with sentiment information to improve patient monitoring.

2. A data model that encodes both diabetic and emotional text features, enabling joint analysis of health and sentiment.

The remainder of the paper is organized as follows. Section 2 reviews related work on diabetes monitoring and health-related sentiment analysis. Section 3 presents the proposed methodology. Section 4 describes the data and experimental results. Finally, Section 5 concludes the paper and outlines future directions.

2 Literature Review

This part presents an extensive examination of the current pieces of writing that are pertinent to the Sentiment Analysis Using Diabetic and Behavioral Data. The review has been organized around two major themes: the first one is the use of AI for the analysis of diabetic data, and the second is the employment of sentiment analysis to interpret patient feedback, with a focus on healthcare contexts.

Advances in AI for Analysis of Diabetic Data Most of the recent AI applications in diabetology has concentrated on the predictive performance of models utilizing structured clinical data. Lee*et. al.,*[6] point out that XGBoost and Random Forest were used for diabetes prediction in older adults, but patient-reported outcomes were not considered, thus, a gap in holistic care remains. Corrao *et. al.,*[4] and Khokhar *et. al.,* [8] conducted the comprehensive reviews, which pointed out that ML/DL models are efficient in analyzing EHR and CGM data, but they hardly provide the translation of complex predictions into the engaging and actionable feedback. In same way, Langarica *et. al.,* [7] and Prioleau *et. al.,* [9] concentrated on algorithmic robustness and dataset creation for glucose forecasting. These researchers primarily focus on achieving technical accuracy while overlooking the importance of presenting understandable insights to the patient, thus, they have pinpointed a significant demand for this work.

Diabetic Sentiment Analysis Sentiment analysis identifies subjective patient experiences, but this information is seldom used in clinical workflows. Feizollah *et. al.,* [10] and Villanueva-Miranda *et. al.,* [1] reviewed health-related NLP and pointed out that the main challenge is to integrate sentiment extraction with behavioral data for personalized interventions. Alkhnbashi *et. al.,* [11] only showed that LLMs such as ChatGPT and DeBERTa could be used on patient reviews without the study linking sentiments to health outcomes. Ossai and Wickramasinghe [12] utilized E-DNNs for app reviews; however, their work was only a snapshot and did not engage with real-time metrics. Naveed *et. al.,* [13] through lexicon-based methods, have established that diabetes affects the mood, but they have also acknowledged that the results are ambiguous unless there are connections with clinical variables.
The recent work, which are outlined in Table 1, show contrasting results: AI models are capable of giving precise instantaneous risk assessments but lack any

emotional context, whereas sentiment analysis can capture the distress of the patients but cannot be aligned with physiological data in time. The proposed research fills the gap that has been identified by suggesting a justified combination of clinical data and sentiment.

Table 1. Comparison of Different State-Of-The-Art solutions.

Reference	Year	Approach	Dataset	Limitations
Subsection A: Advances in AI for Analysis of Diabetic Data				
Lee et. al., [6]	2025	ML (XGBoost, Random Forest)	KNHANES 2019-2021 (4,917 adults)	Relies only on structured clinical data, ignoring patient-reported outcomes.
Corrao et. al., [4]	2025	Review of ML/DL methods	Various (EHRs, CGM data)	Fails to translate predictions into engaging feedback.
Langarica et. al., [7]	2024	DL (LSTM, Encoder-Decoder)	Curated dataset	Focuses on accuracy; lacks motivating presentation for patients.
Khokhar et. al., [8]	2025	Systematic Review	53 studies	Emphasizes data/explainability; ignores user engagement.
Prioleau et. al., [9]	2025	Data Compilation	Glucose-ML (10 CGM datasets)	Focuses on benchmarks, not patient interaction.
Subsection B: Diabetic Sentiment Analysis				
Feizollah et. al., [10]	2025	Scoping Review (NLP)	Reviews, surveys, social media	Needs integration of sentiment with behavioral data.
Alkhnbashi et. al., [11]	2024	LLM (ChatGPT), DeBERTa	Patient reviews	Does not link sentiments to health outcomes.
Villanueva-Miranda et. al., [1]	2025	Systematic Review	Social media, forums	Lacks personalized interventions based on sentiment.
Ossai & Wickramasinghe [12]	2023	E-DNN, LDA	38k app comments	Static analysis; not linked to real-time behavioral data.
Naveed et. al., [13]	2024	TextBlob, Vader	215 forum posts	Mood effects inconclusive; not tied to clinical outcomes.

3 Proposed methodology

Sentiment analysis, one of the major NLP techniques, is capable of identifying the feeling or the side of the argument of a given text by means of either lexicon-based or deep learning models like BERT and DeBERTa. On the other hand, in a similar fashion, AI methods such as LSTM, XGBoost, and CNN–GRU have brought about the rise of diabetes prediction accuracies when the predictions are based on physiological data. The catch is, these techniques function separately and there is no bridge between them in terms of integration of emotional and clinical indicators. The merger of these two sources leads to the complete understanding of the wellness of diabetic patients. Conventional systems have been designed based on numerical metrics like glucose, HbA1c, and BMI and have been failing in psychological and behavioral factors, which have a strong influence on the commitment and the results.

Problem Scenario:A diabetic patient may record stable glucose readings but still experience negative emotions like stress or fatigue that reduce compliance with treatment. Conventional analytics fail to detect such emotional fluctuations. **Problem Definition:** To design an intelligent sentiment-driven framework that integrates behavioral and physiological data to predict the patient's emotional well-being and provide adaptive insights.

Mathematical Model: Let

$$D = \{d_1, d_2, \ldots, d_n\}, \quad S = \{s_1, s_2, \ldots, s_m\}$$

represent diabetes parameters and sentiment features respectively. The fusion function $F(D, S)$ computes a combined emotional–health score:

$$H = F(D, S) = \alpha f_1(D) + \beta f_2(S)$$

where α, β are weights learned by the embedded deep neural network (E-DNN). This work aims to build one system that will comprise physiological data and diabetes-related data fused with sentiment data for better patient monitoring. The proposed model features not only diabetic but also emotional text features, thereby allowing health and sentiment to be analyzed jointly for a more comprehensive evaluation. It is assumed that the textual data from the GoEmotions dataset [15] are the most accurate emotional expressions, and the diabetic dataset [14] is the most reliable source of physiological parameters.

3.1 System Architecture

The system architecture of the proposed system incorporates the integration of sentiment and behavioral data through a carefully planned six-stage pipeline as depicted in Fig. 1 that was designed to represent a shift from data-driven monitoring to emotion-aware health analytics. It starts with the Input Module representing the two fundamental streams: the emotional text data derived from the GoEmotions dataset and the physiological-behavioral data obtained from the Diabetics dataset (IEEE Dataport 2024). The unprocessed data is transferred to the Data Preprocessing Module. In this module, the text is cleaned, the tokens are obtained, and lemmatization is done by NLTK and spaCy, whereas the numerical attributes are normalized, and missing values are filled using Pandas and NumPy. In the next Feature Extraction Module, the words are changed into TF-IDF vectors, and the numeric features like glucose and BMI are rescaled to maintain compatibility. The main analysis is done in the Sentiment Modeling Module that uses an Embedded Deep Neural Network (E-DNN) to differentiate emotional states like happiness, stress, or sadness. The outputted sentiment vectors are finally coupled with the physiological features by the Fusion Layer to produce a single emotional-health embedding.Finally, the Output Layer maps these holistic profiles to avatar expressions, providing visual representations of the patient's state (e.g., fatigued, happy) to enhance engagement and well-being.

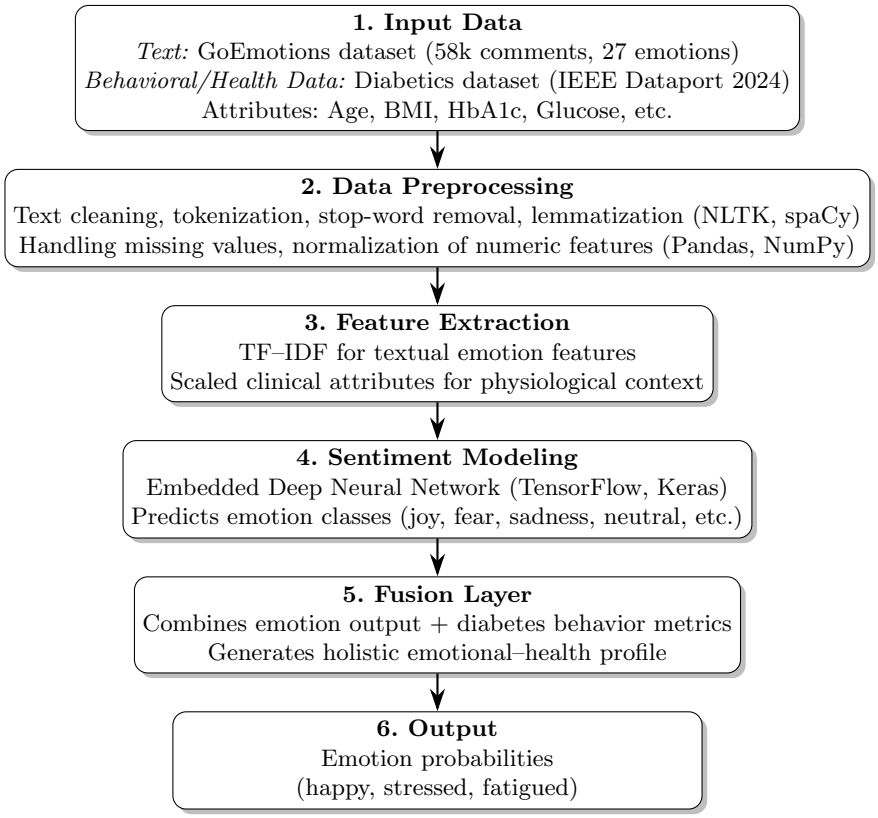

Fig. 1. System architecture integrating textual sentiment with behavioral and clinical diabetes data to produce a fused emotional–health profile.

The model is trained on the GoEmotions dataset [15]. The resulting emotion probability vector from the chosen model serves as the primary input to the Fusion Layer (Module 5). In that module, this sentiment vector is concatenated with the normalized physiological and behavioral data (e.g., Glucose, BMI, HbA1c) from the Diabetics dataset [14] to create the final, unified emotional-health profile. The proposed methodology centers on a six-module system architecture (Fig. 1) designed to fuse emotional and clinical data. The system ingests two distinct data streams: textual data from the GoEmotions dataset [15] and physiological/behavioral data from a diabetics dataset [14]. After preprocessing and feature extraction (using TF-IDF for text), the data enters the Sentiment Modeling module. This module, illustrated in Fig. 2, implements a novel Embedded Deep Neural Network (E-DNN) to produce an emotion probability vector.The principal innovative concept of the work is at the Fusion Layer, the place the sentiment vector is joined with the normalized clinical features to create a single, unified emotional-health profile for deep reasoning.

4 Results and Discussion

Every experimental scenario is carried out with Python 3.9, TensorFlow, Keras, and Scikit-learn in a Kaggle setting which is powered by dual NVIDIA T4 GPUs. The model learns from a filtered GoEmotions dataset [15] containing 128,664 samples and a Diabetic based dataset [14].
The proposed E-DNN model is evaluated on an enhanced test set of 20 samples, specifically curated to include diabetes-related scenarios (e.g., "My blood sugar is critically low"). The model's predictions are shown in Table 2.
Table 3 illustrates the metrics of the performance. The model was able to reach an overall accuracy of 75.00%. Essentially the model was able to discriminate very well between the extreme, single-peak emotions such as 'anger' and 'fear' (very important for the detection of patient distress), thus practically achieving an f1-score of 1.00. The problem was, however, that the model had difficulty with subtle differentiated categories such as 'approval' (0.00 f1-score), and it was most times that the category 'neutral' or 'gratitude' was mislabeled as 'approval'.
To further validate the model's robustness, A detailed performance analysis is conducted using training history and text similarity metrics. Fig. 3 illustrates the E-DNN's learning progress over 20 epochs. The Training vs. Validation Loss (Fig. 3) shows a rapid decrease in training loss, though the validation loss begins to diverge after epoch 5, indicating potential overfitting. However, the Classification Error plot (Fig. 4) confirms that the error rate stabilized around 0.45, demonstrating that the model effectively learned the core features of the dataset despite the noise inherent in social media text.

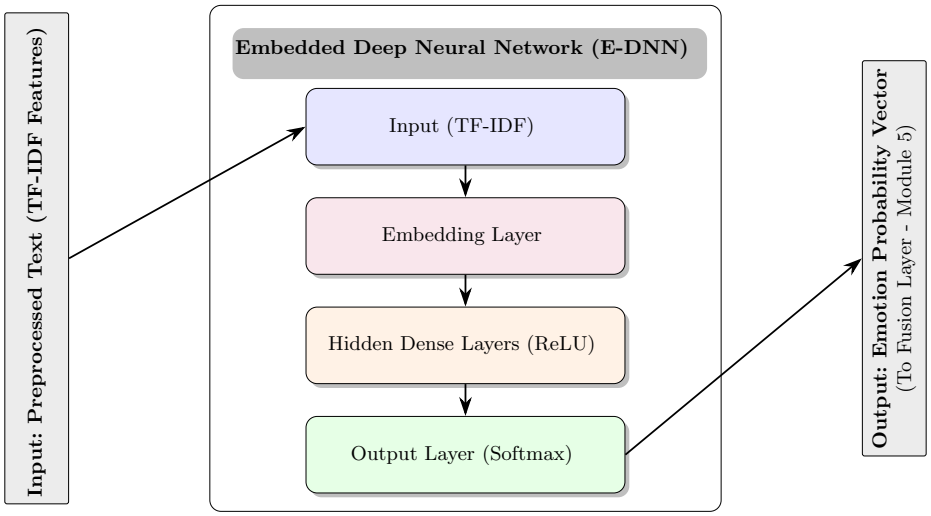

Fig. 2. Detailed architecture of the proposed Embedded Deep Neural Network (E-DNN) for Sentiment Modeling (Module 4).

Table 2. Test Set Predictions (E-DNN Model) - Selected Samples

Idx	Text (Excerpt)	True	Pred
0	I love this movie so much! It's absolutely amazing...	joy	joy
1	This is making me really angry and frustrated...	anger	anger
	... (rows 2–11 omitted for brevity) ...		
12	My blood sugar is critically low and I'm terrified...	fear	fear
13	I am so frustrated that my glucose is high despite...	anger	anger
14	Finally got my A1C down to 6.5! I am absolutely thrilled!	joy	approval
15	I hate having to inject insulin every single day...	sadness	anger
16	The doctor said my progress is excellent. I'm relieved.	approval	neutral

Table 3. Classification Report: E-DNN (Proposed Model)

	precision	**recall**	**f1-score**	**support**
anger	0.67	1.00	0.80	2
approval	0.50	0.50	0.50	2
confusion	1.00	0.50	0.67	2
disappointment	0.00	0.00	0.00	1
disgust	1.00	1.00	1.00	1
excitement	1.00	1.00	1.00	1
fear	1.00	1.00	1.00	3
gratitude	1.00	1.00	1.00	1
joy	1.00	0.50	0.67	2
neutral	0.40	1.00	0.57	2
sadness	1.00	0.50	0.67	2
surprise	1.00	1.00	1.00	1
accuracy			**0.75**	20
macro avg	0.80	0.75	0.74	20
weighted avg	0.81	0.75	0.74	20

To validate semantic robustness, the alignment between predicted and true labels using BLEU and ROUGE metrics are evaluated. BLEU calculates precision-based n-gram overlap penalized by brevity:

$$BLEU = BP \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right) \tag{1}$$

Conversely, ROUGE-N measures recall-based n-gram overlap:

$$ROUGE\text{-}N = \frac{\sum_{S \in \{Ref\}} \sum_{gram_n \in S} \text{count}_{match}(gram_n)}{\sum_{S \in \{Ref\}} \sum_{gram_n \in S} \text{count}(gram_n)} \tag{2}$$

The E-DNN was able to get an Average BLEU of 0.7376 and ROUGE-1/L of 0.7500. These figures reflect that the incorrect classifications still have a strong semantic closeness to the actual data (for instance, 'joy' for 'relief'), thus, supporting the model's contextual effectiveness.

Fig. 3. Epoch vs. Loss

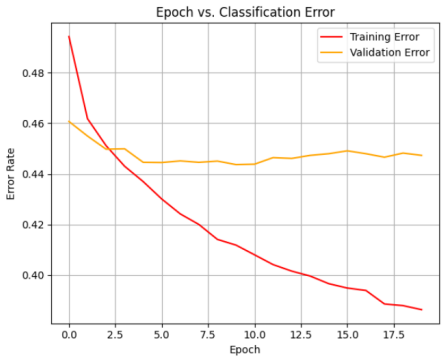

Fig. 4. Epoch vs. Classification Error

5 Conclusion

The work overcame one of the biggest challenges in diabetes care by addressing the lack of integration between physiological monitoring and psychological assessment through the proposal of a new integrative framework. In particular, the team has effectively proven the possibility of combining these data sources with a Deep Neural Network that was Embedded (E-DNN). On the test set specific for diabetes, the E-DNN sentiment modeling module reached a fairly good 75.00 percent overall accuracy. Although the model exhibited excellent capacities in recognizing situations with high emotional impact such as anger and fear (F1-score 1.00), it had difficulty with the subtlety of emotions, for instance, approval. Yet, the elevated BLEU (0.7376) and ROUGE (0.7500) scores are indicative that the model's output is in line with the strong semantic norms. The major point of this study is, in fact, the integrative architecture which not only resolves the problem of data silos but also addresses the issue of emotional states by linking them directly to clinical data. Subsequent research will be directed towards improving the sentiment model with contextual embeddings (e.g., BERT) to capture emotional nuance better and implementing the system for actual-time patient communication through the suggested avatar interface.

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